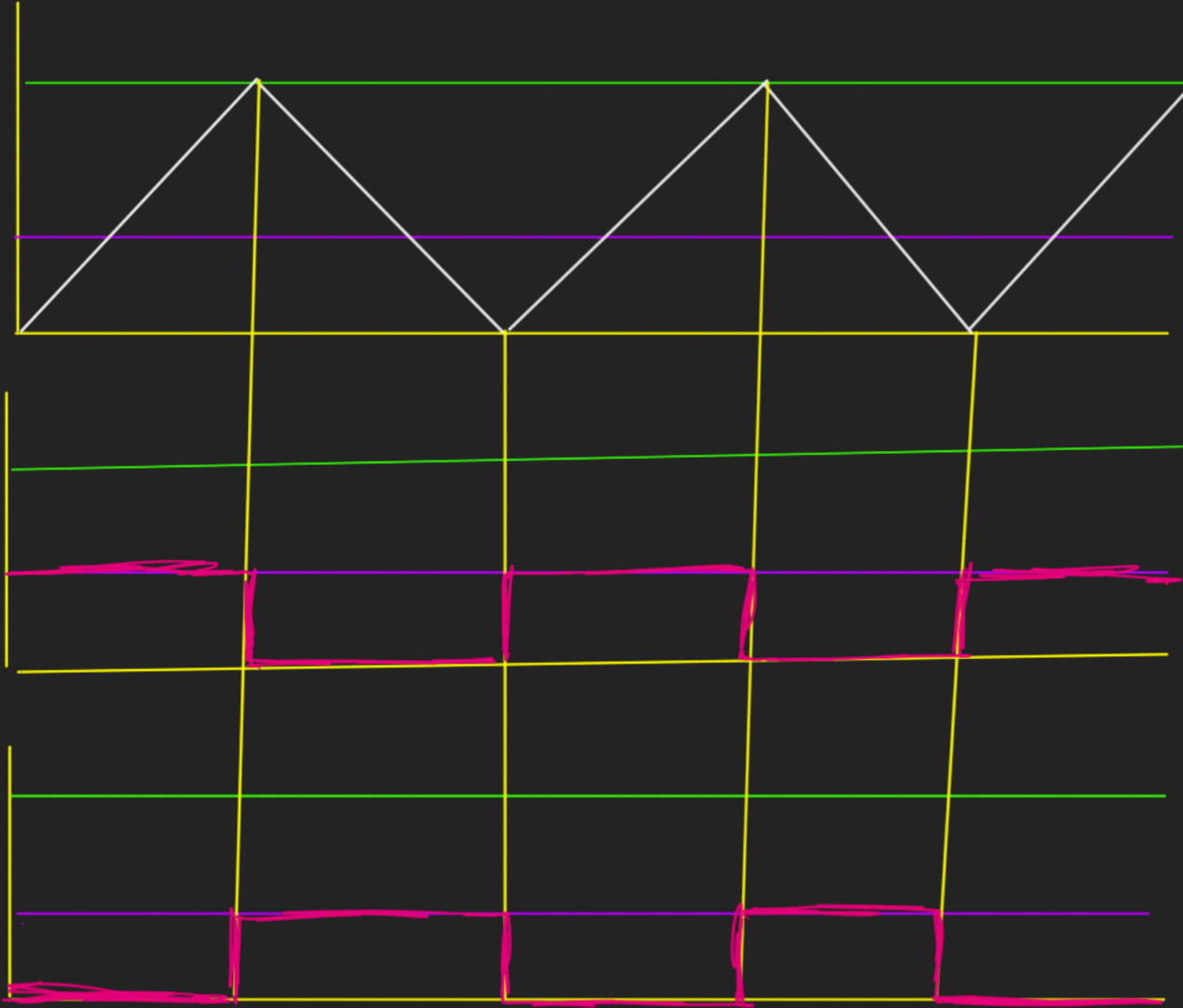


# \* Timer / Phase Correct Mode



OCRO = 100

[1] clear

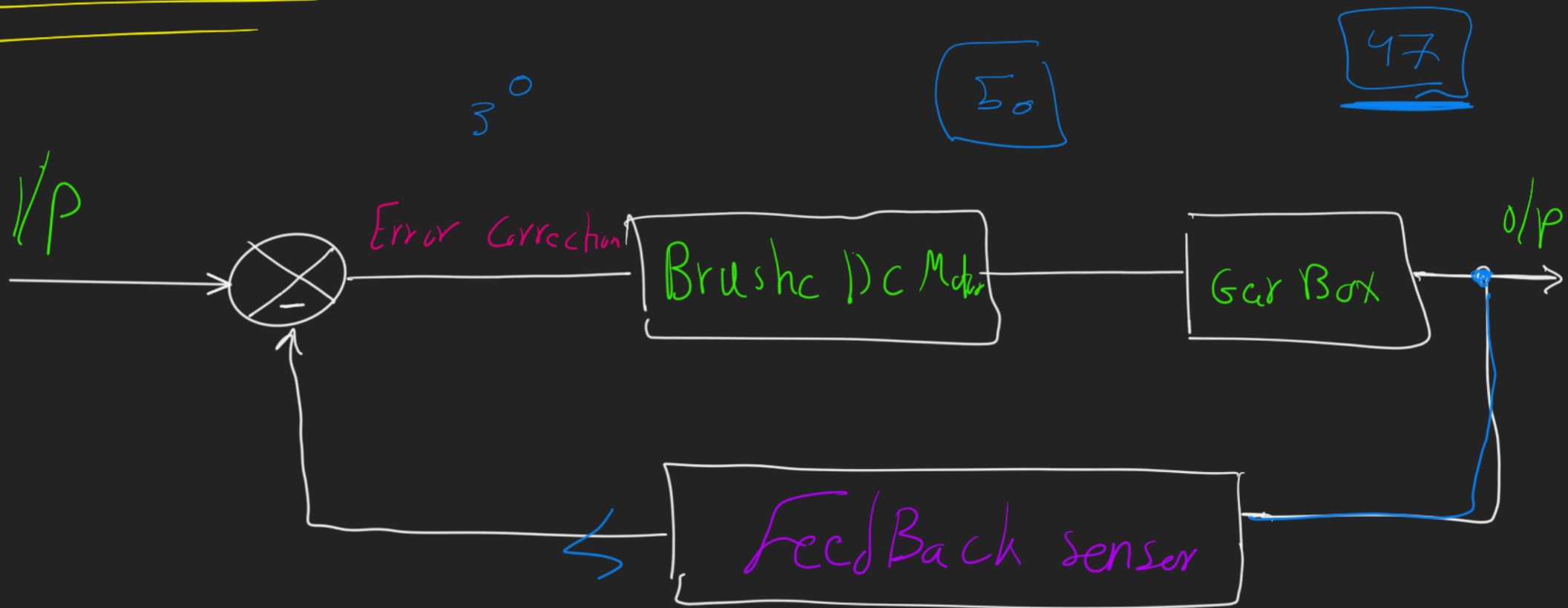
Set uCOPIN

[2] set

clear uCOPIN

Driver of phase correct on you

Servo Motor → Closed Loop systems



Σ<sub>o</sub> → Correction, closed loop system

# Operate Servo Motor

PWM Freq = 50 Hz

Amplitude  $\rightarrow$  5V

Duty cycle

X

8 Bits

16-Bit timer

1  $\rightarrow$  2



0

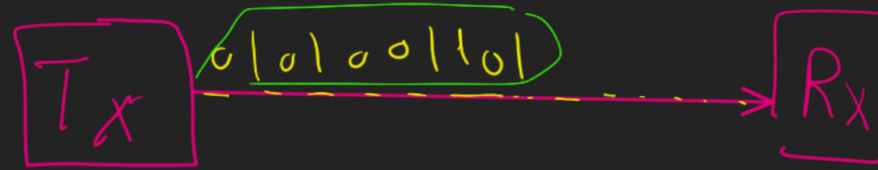


180

# \* Communication protocols

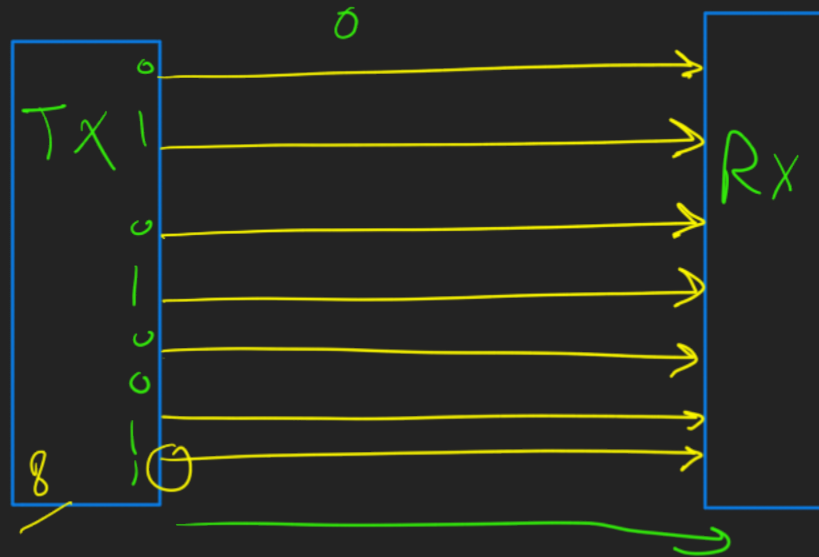
I

## \* Serial



|          | Serial | Parallel |
|----------|--------|----------|
| Speed    |        | ✓        |
| Security | ✓      |          |
| Safety   | ✓      |          |
| Cost     |        | ✓        |

## \* parallel



# \* $T_x$ $R_x$ Relation

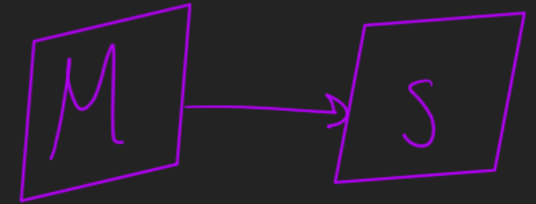
[2]

## [1] Peer to Peer

انتبه قايده على قهوة لود

## [2] Master - slave

→ Single Master, single slave



→ " " , Multiple slaves

زى السبكه

→ Multiple Masters, Multiple slaves

زى الكليه

→ Multiple Masters, No slaves

مركزا عن شركه

\* Synchronization ~ جزا 3

↳ clk is shared

\* Asynchronous

↳ clk is Not shared, No clk.

→ Baud Rate (#)

# \* Data Direction

4

[1] simplex



[2] Half Duplex

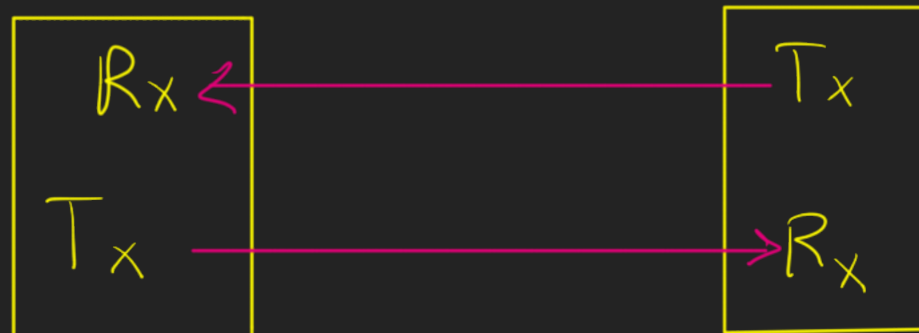


[3] Full Duplex



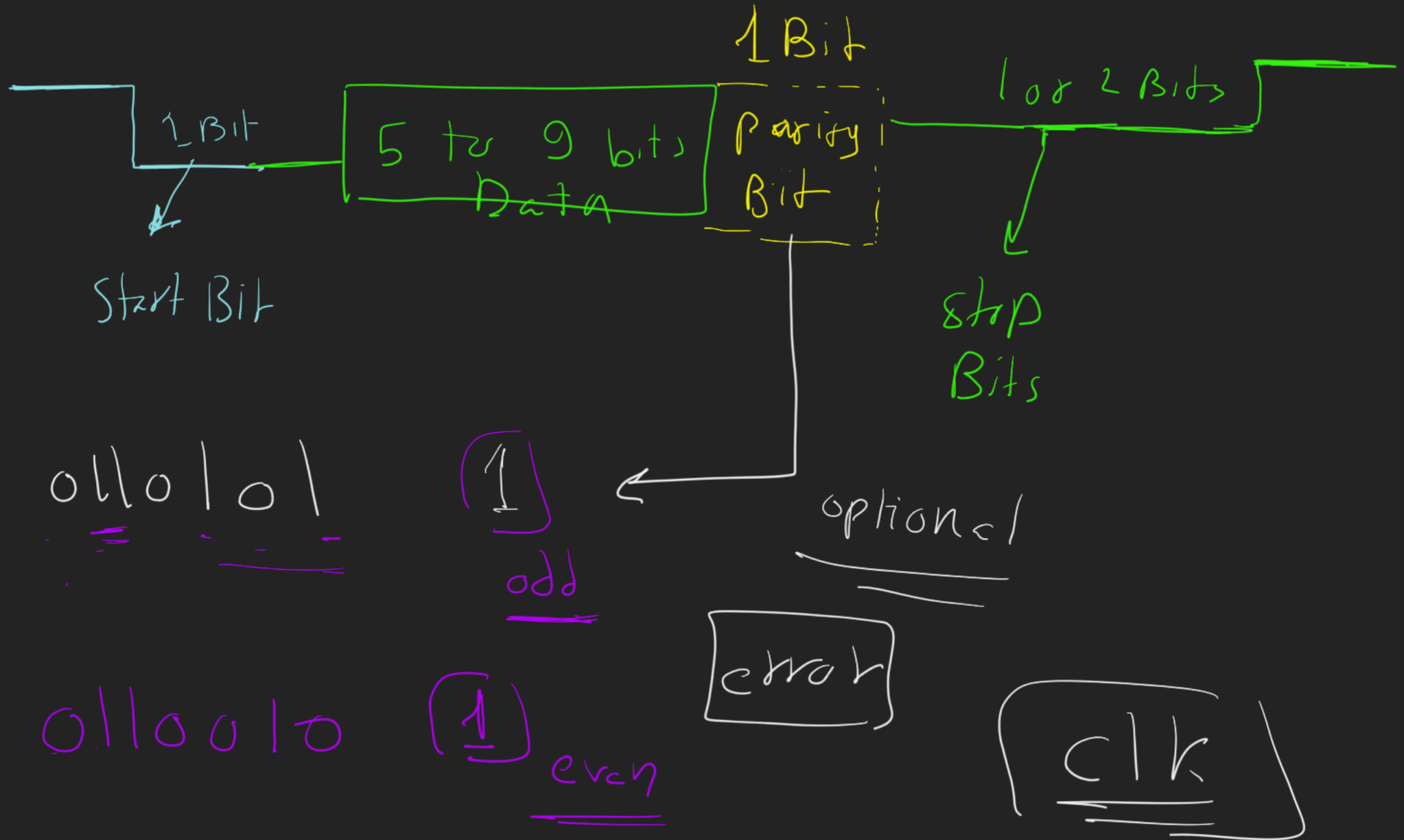
\* UART  $\xrightarrow{\quad}$  Transmitter  
           $\downarrow$        $\xrightarrow{\quad}$  Receiver  
          Universal      Asynchronous

- [1] wired    [2] serial    [3] peer to peer  
[4] Async    [5] half / Full Duplex





# (3) Data Frame



Baud Rate / clk

9600

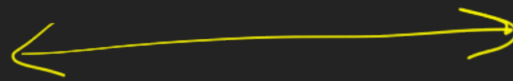
115200

1st Device

115200

2nd Device

9600



Serial Monitor

Case

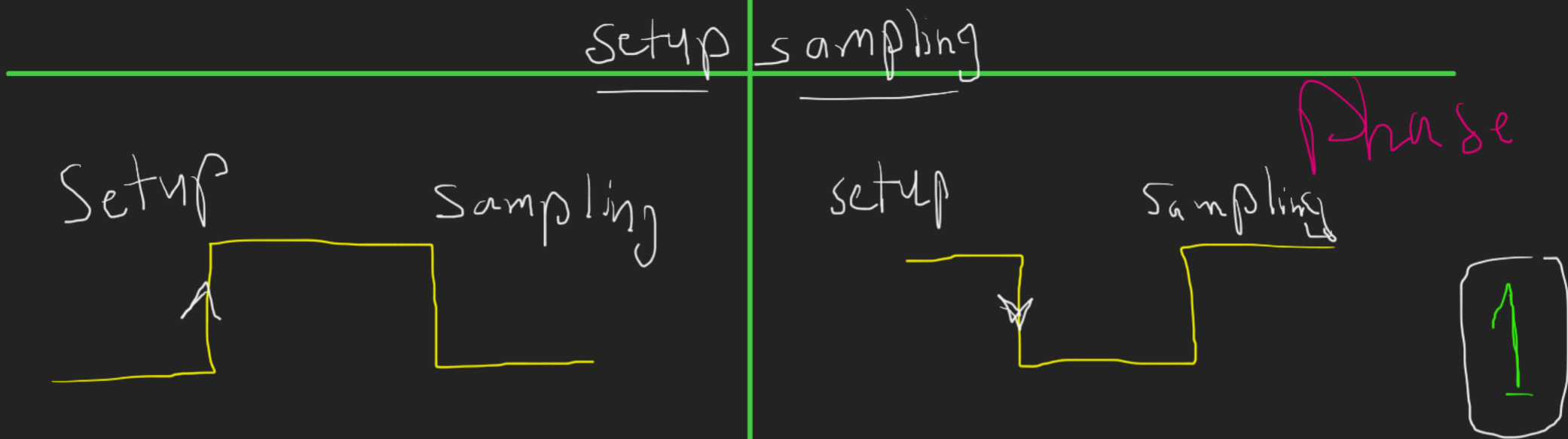
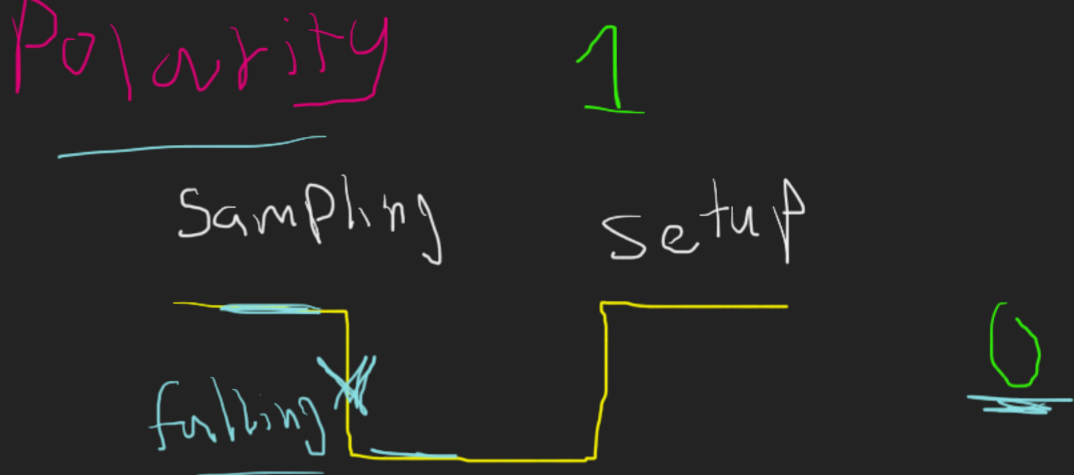
9600

دیفولت

# SPI → Serial Peripheral Interface

- ① wired
- ② Serial
- ③ Full Duplex
- ④ sync
- ⑤ Single Master, Multi slaves





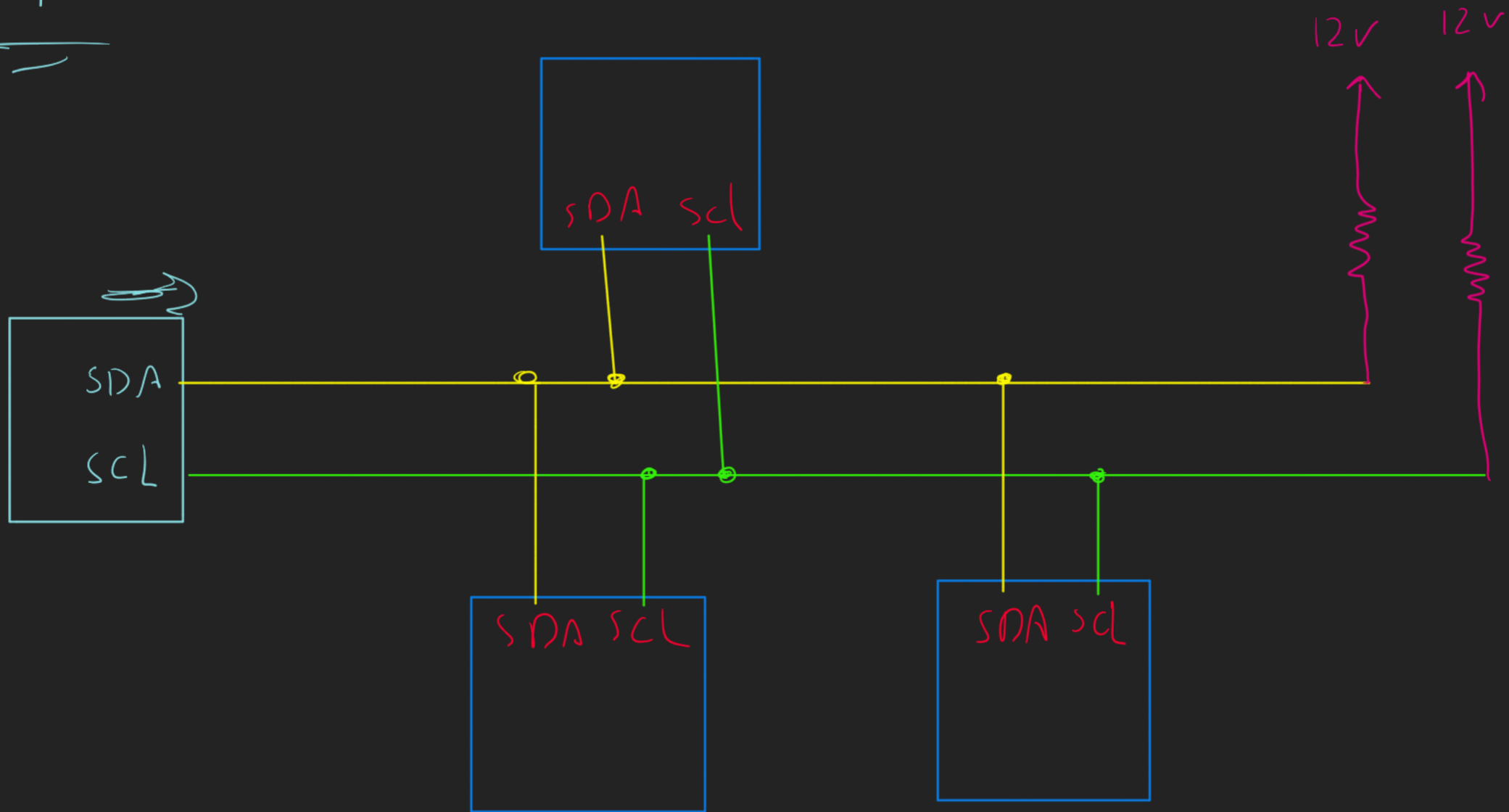
I2C Inter Integrated circuit

TwI Two wire Interface

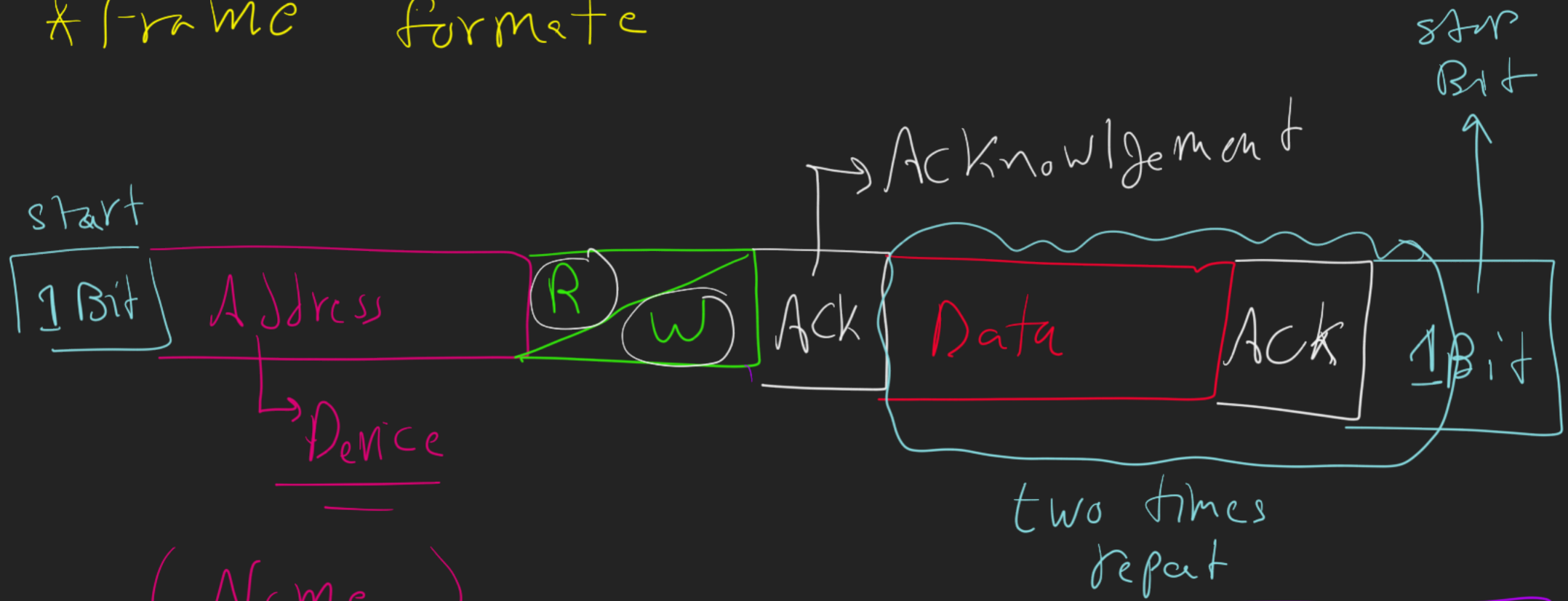
[1] wired [2] sync [3] serial [4] Half Duplex  
[5] Multi Masters, Multi slaves



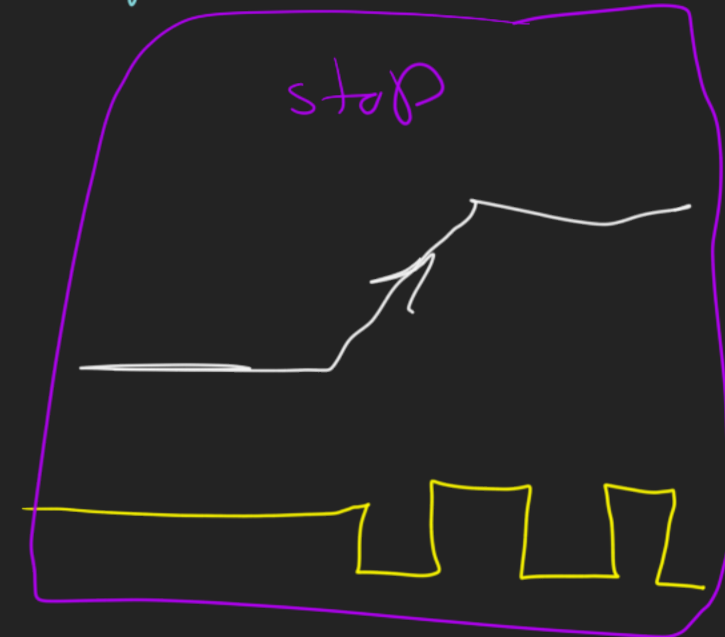
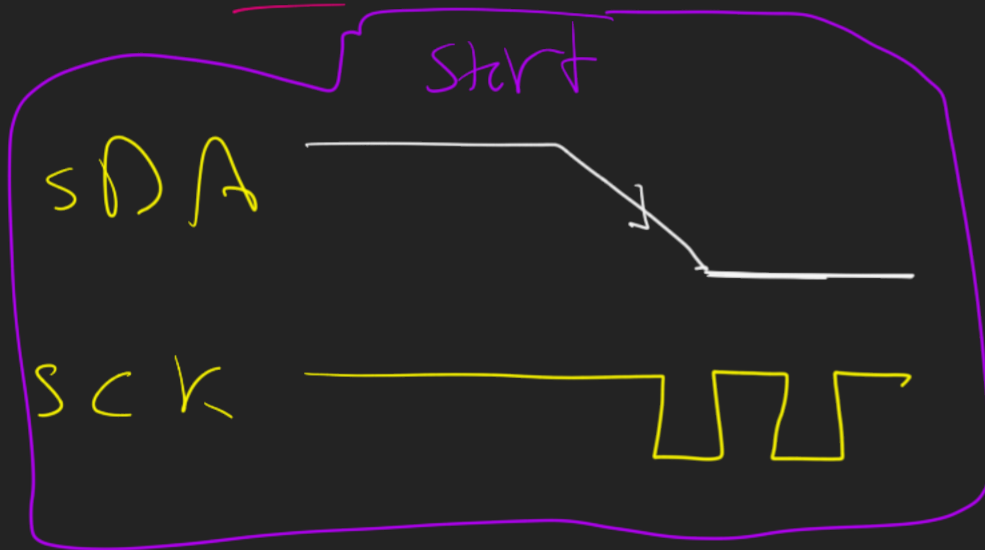
HW



# \* Frame format



( Name )



up to 129 Device theory

112 practical

# problem

too slow



